IMPLEMENTATION STATUS REPORT

Gap-Closure Update

Implementation status of the 17 registered items from "Gap Analysis: ca.pdf vs Implemented System" after the gap-closure phase

> Baseline report: Gap Analysis, 12 November 2026 (17 items; 4 High)
>
> System reviewed: Next.js 16 + Prisma platform, now 34 data models
>
> Evidence: acceptance suite extended from 73 to 134 checks — all passing
>
> Date: 12 November 2026

Table of Contents

1. Executive Summary.....

2. Scope, Method and Evidence.....

3. Item-by-Item Status.....

4. Test Evidence.....

5. What Still Stands Between the Platform and Client Onboarding.....

Note: This Table of Contents is generated via field codes. To ensure page number accuracy after editing, please right-click the TOC and select "Update Field."

1. Executive Summary

The gap analysis dated 12 November 2026 compared the concept document ca.pdf against the implemented platform and registered 17 mismatch items: 5 missing, 10 partial and 2 by-design deviations, of which 4 were rated High priority for correction before client onboarding. This update reports what happened after that review: a dedicated gap-closure phase implemented the closed items as working software and extended the automated acceptance suite to prove them.

Of the 17 items, 13 are now closed: all four High-severity gaps (R-02 period linkage, R-03 reminder engine, R-04 verified statutory due dates, R-06 client tax payment) and nine of the ten Medium/Low gaps (R-01, R-05, R-08, R-09, R-10, R-11, R-12, R-13, R-14). One item remains an open business decision (R-15 multi-firm), one remains externally blocked pending provider credentials (R-07 notification channels), and the two by-design deviations are unchanged by intent.

The acceptance suite grew from 73 to 134 automated checks, one per behaviour described in this report, and all pass against a freshly seeded server. The data model grew from 22 to 34 Prisma models. Documentation (architecture, setup, CA review checklist) has been updated in step. The platform's core discipline is unchanged: proactive date discipline built on verified dates, period-scoped cases, and honest filing semantics.

Table 1: Registered items — before and after the gap-closure phase

Original status	Items	Status after gap closure
Missing (5)	R-02, R-03, R-06, R-07, R-11	3 closed (R-02, R-03, R-06); R-07 remains externally blocked (channel credentials); R-11 closed
Partial (10)	R-01, R-04, R-05, R-08, R-09, R-10, R-12, R-13, R-14, R-15	9 closed (R-01, R-04, R-05, R-08, R-09, R-10, R-12, R-13, R-14); R-15 remains an open business decision
By-design deviation (2)	R-16, R-17	Unchanged — intentional and endorsed by ca.pdf's own safeguards

2. Scope, Method and Evidence

Every closure was implemented in the same repository and verified at three levels. First, schema: new master and transaction tables (StatutoryDueDate, ReminderPolicy, ReminderLog, TaxPayment, RegulatoryUpdate, ConsentRecord) and new fields (Application period columns, Business.annualTurnover) were added to the Prisma schema and pushed to the database. Second, server: obligation generation, draft creation, dashboards and the reminder sweep enforce the new behaviour server-side — eligibility, period validation and tax-payment gating are not merely hidden in the UI. Third, tests: each closure has a dedicated acceptance suite that runs against the live API and, for encryption, against the bytes actually written to disk.

The suite is designed to run on a clean seed (bun scripts/seed.ts, then bun scripts/acceptance-tests.ts). At the time of writing: 134 passed, 0 failed, and the TypeScript build is clean. Where a closure depends on external facts — statutory dates, eligibility thresholds — the implementation stores configuration with source provenance and a demo-marked flag; it deliberately does not invent legal values. That honesty requirement from the original gap analysis carries through every new feature.

3. Item-by-Item Status

The register below restates each item from the gap analysis with its after-state and where the implementation now lives. PDF references and the original severity are unchanged from the first report.

Table 2: Gap-closure register (17 items)

ID	Requirement (ca.pdf)	Before	After	How it is now implemented
R-01	Dedicated Knowledge / Rule Manager role	Partial	Closed	KNOWLEDGE_MANAGER role added. Maintains rule masters (statutory due dates, reminder policies, regulatory updates) and rule content of configuration drafts (form schema, document requirements, task templates, applicability, compliance rules, sources). Pricing and workflow wiring stay admin-only; professional approval stays CA-only; organizational masters stay admin-only. UI limits the Rule Masters view to the rule tabs. Seeded demo account knowledge@demo.in.
R-02	Case linked to a chosen period	Missing	Closed	Application carries periodType/periodLabel/periodStart/periodEnd. The apply wizard adds a statutory-period step (assessment-year or tax-period options per the service's periodMode); resolvePeriod() validates labels server-side and recomputes bounds so they can never disagree. Generated obligations and filings inherit the case period instead of a generic current-month label.

ID	Requirement (ca.pdf)	Before	After	How it is now implemented
R-03	Configurable reminder & escalation policy	Missing	Closed	ReminderPolicy master (firm-wide default or per-service; appliesTo COMPLIANCE / INFO_REQUEST; early intervals JSON e.g. [15,7,2]; repeat cadence; escalation delay + target role). A scheduled sweep (instrumentation hook + POST /api/reminders/run) walks open obligations and pending information requests and creates deduplicated EARLY / DUE / OVERDUE / ESCALATION / INFO_REQUEST notifications, each logged in ReminderLog. Demo-marked dates are disclosed in reminder copy.
R-04	Due-date rules as verified statutory dates	Partial	Closed	StatutoryDueDate master: per service code + obligation code + period label, with due date, source-reference provenance, effectiveFrom/To, extension flag + note, and a verification trail (verifiedBy/at). Verification is refused without a source. Obligation generation prefers master dates over demo offsets and stamps the obligation's note with the date's provenance (verified vs demo-marked).
R-05	Official-source monitoring	Partial	Closed	RegulatoryUpdate intake: staff report updates against registered sources with affected services; verification restricted to CA / admin / knowledge manager; verification computes an impact snapshot over frozen configuration snapshots; only verified updates can be marked applied. Manual first — no scraping — consistent with the platform's safeguards.
R-06	Client's tax payment tracked per case	Missing	Closed	TaxPayment records on the application: liability type (self-assessment, advance tax, TDS, GST cash, other), period, amount, due date, status (PENDING / PAID / WAIVED / NOT_APPLICABLE). Marking PAID requires a challan reference; staff record, client views; the readiness computation blocks on pending tax.
R-07	Email / portal / WhatsApp channels	Missing	Open	Externally blocked by design: in-app remains the source of record; delivery adapters need provider credentials and consent records (now in place via R-11). No fake "email sent" states.
R-08	Combined readiness check per deadline	Partial	Closed	computeReadiness() joins documents, professional fee, client tax and open work per upcoming obligation into READY / AT_RISK / BLOCKED with reasons; surfaced on client and admin dashboards' upcoming-deadline lists.
R-09	Admin workload & reassignment overview	Partial	Closed	Team Workload report: per-employee open tasks and utilization, department aggregation, and reassignment history fed from TaskHistory; clients are refused (403).
R-10	Document protection: encryption & scanning	Partial	Closed	Every stored document is sealed with AES-256-GCM before touching disk (ENC1: framing; key from FILE_ENCRYPTION_KEY or auto-provisioned 0600 key file; transparent legacy read-back). Upload content screening rejects EICAR signatures, executable magic bytes, shebang scripts and MIME/content mismatches before acceptance. Full antivirus engines remain a documented production hardening item.
R-11	Client consent & authorization records	Missing	Closed	ConsentRecord with scope (e.g. DATA_RETRIEVAL_AIS), purpose, grant/revoke, expiry, evidence and granted-by role; client portal screen; staff visibility. Foundation required before any Stage-6 integration work.

ID	Requirement (ca.pdf)	Before	After	How it is now implemented
R-12	Identify cases affected by a rule change	Partial	Closed	Covered by the regulatory-update impact snapshot: verification computes affected applications from frozen config snapshots and records reference numbers, giving a guided impact-to-apply workflow.
R-13	Applicability over client attributes	Partial	Closed	Shared applicability evaluator: requires (pan/gstin/tan/cin each YES/NO/ANY) plus minTurnover/maxTurnover over client-declared Business.annualTurnover. Enforced server-side at draft creation (422 NOT_ELIGIBLE with reasons); the wizard filters ineligible profiles and shows the human-readable reasons per business. Thresholds are configuration facts flagged for professional verification, never invented statutory values.
R-14	Individual taxpayers without a business entity	Partial	Closed	businessId is optional; applicantType applicability (BUSINESS / INDIVIDUAL / ANY); seeded ITR reference service is individual-only with ASSESSMENT_YEAR period mode; wizard files without a business profile.
R-15	Single firm or multi-firm platform	Partial	Open	Unchanged — remains an architecture decision for the firm; documented as the largest structural unknown.
R-16	Authorized retrieval of client records (AIS, 26AS)	By-design	Same	Not implemented intentionally; consent records (R-11) are the necessary foundation and now exist.
R-17	Stage 6: filing/payment integrations	By-design	Same	Intentionally out of scope for this release.

4. Test Evidence

The automated suite is the primary evidence for every closure claim in this report. The table maps each suite to the behaviours it locks in, so a reviewer can re-run any claim independently. The full suite completes against a freshly seeded server in a few minutes.

Table 3: Acceptance suites added or extended during gap closure

Suite	What it proves
T1–T15 (master prompt §26)	Conditional documents, constitution switching, circular-rule rejection, snapshot preservation, duplicate-task prevention, assignment fallback, record-level authorization, document rejection, webhook idempotency, income-tax framework note, Udyam zero fee, filing evidence gates, dependency unlocking, publish/approval order — plus the full happy path to completion and compliance generation.
R-02 / R-14 suite	ITR service individual-only, ASSESSMENT_YEAR mode, draft without period rejected, malformed AY label rejected, draft carries the chosen period, no business entity attached.
R-06 suite	PAID without challan reference rejected, pending liability recorded, paid accepted with challan, client sees own records, staff-only recording.
R-04 suite	Seeded due-date rows with sources, demo-marked until verified, verification without source rejected, verified rows record the verifier, duplicate period rejected.

Suite	What it proves
R-03 suite	Policies seeded active, empty intervals rejected, create/deactivate policy, sweep runs with deduplicated output.
R-11 suite	Client grants and revokes scoped consents; records carry granted-by role.
R-05 / R-12 suite	Report against registered source, employee verification refused, CA verification computes impact snapshot, only verified updates can be applied.
R-09 / R-08 suite	Workload aggregates per employee/department with reassignment history; dashboard exposes per-deadline readiness; case-linked deadline BLOCKED by pending tax payment.
R-13 suite	Service with requires GSTIN + turnover bounds published through the approval gates; draft blocked server-side without GSTIN; allowed once registered and within bounds; blocked above the turnover limit.
R-01 suite	Knowledge manager opens drafts and maintains rule content; cannot change pricing, cannot approve versions, cannot create organizational masters or list employees; can manage due dates and regulatory updates.
R-10 suite	EICAR and executable uploads rejected; valid upload round-trips byte-identical; new files on disk are ENC1:-sealed with no plaintext written; another client cannot download the document.

5. What Still Stands Between the Platform and Client Onboarding

The remaining work is deliberately outside the codebase. First, professional verification: a qualified CA must verify the seeded service configurations, the statutory due-date master rows and any eligibility thresholds against current official sources before they are relied on for real filings; the CA review checklist has been extended with dedicated sections for due dates, eligibility thresholds and reminder policies. Second, notification channels: email, SMS or WhatsApp delivery requires provider credentials and a licensed business account; the in-app centre remains the source of record and no delivery is faked. Third, the architecture decision on single-firm versus multi-firm operation (R-15) should be taken before significant onboarding, because retrofitting tenancy later is costly. Fourth, production hardening as documented in the setup guide: an explicit FILE_ENCRYPTION_KEY backed up separately from the database, a system cron or worker for the reminder sweep, PostgreSQL migration, and error monitoring.